DSC 102 — Spring 2026

# Midterm Review — Set A

# Problem 1

You are working with an 8-bit signed two's complement system.

Compute: $95 + 60$

1. Does overflow occur?
2. What 8-bit result does the machine produce?
3. What decimal value does that bit pattern represent?

## Solution 1

1. Yes. $95 + 60 = 155$, which is larger than the 8-bit signed maximum of $127$.
2. $155_{10} = \mathbf{10011011_2}$
3. `10011011` represents $-\mathbf{101}$ in signed two's complement.

# Problem 2

A memory address is given as:

`0x3ACF`

Questions:

1. Convert it to binary.
2. How many bytes long is this address?
3. If each address points to 1 byte, what is the maximum addressable memory?

## Solution 2

1. `0x3ACF = 0011 1010 1100 1111`
2. 4 hex digits = 16 bits = $\mathbf{2}$ bytes
3. $2^{16} = 65{,}536$ bytes $= \mathbf{64}\ \mathrm{KB}$

# Problem 3

In a RISC-like system:

- Each instruction is 8 bytes
- Instructions must be 8-byte aligned
- PC starts at `0x4000`
- Instruction memory size is 2 KB

Questions:

1. What are the next 5 instruction addresses?
2. How many instructions fit in 2 KB?
3. What fraction of all 32-bit byte addresses are valid instruction addresses?

## Solution 3

1. `0x4008`, `0x4010`, `0x4018`, `0x4020`, `0x4028`
2. $2048/8 = \mathbf{256}$ instructions
3. Only addresses with low 3 bits `000` are valid, so the fraction is $\mathbf{1/8}$.

# Problem 4

A processor runs at 4 GHz. A program has:

- 50% arithmetic instructions, 1 cycle each
- 20% control instructions, 3 cycles each
- 30% memory instructions, 2 base cycles each
- Each memory instruction also stalls for 40 cycles
- Total instructions = $1.5 \times 10^9$

Questions:

1. What is the average CPI without stalls?
2. What is the average CPI including stalls?
3. What is the total execution time?
4. If cache reduces stall from 40 cycles to 10 cycles, what is the speedup?

# Solution 4

1. $0.5(1) + 0.2(3) + 0.3(2) = \mathbf{1.7}$
2. $1.7 + 0.3(40) = \mathbf{13.7}$
3. Cycles $= 1.5 \times 10^9 \cdot 13.7 = 2.055 \times 10^{10}$, so

$$t = \frac{2.055 \times 10^{10}}{4 \times 10^9} = \mathbf{5.14}\ \mathrm{s}$$

4. New CPI $= 1.7 + 0.3(10) = 4.7$, so

$$\text{speedup} = \frac{13.7}{4.7} = \mathbf{2.91}\times$$

# Problem 5

A system has:

- Cache access time = 8 ns
- DRAM access time = 120 ns
- Cache hit rate = 0.75

Questions:

1. What is the AMAT?
2. If hit rate improves to 0.9, what is the new AMAT?
3. What is the speedup?

## Solution 5

1. $0.75(8) + 0.25(120) = \mathbf{36}\ \mathrm{ns}$
2. $0.9(8) + 0.1(120) = \mathbf{19.2}\ \mathrm{ns}$
3. $\dfrac{36}{19.2} = \mathbf{1.875}\times$

# Problem 6

Assume:

- Cache line size = 64 bytes
- Each `float64` = 8 bytes
- Matrix stored in row-major order

Consider:

```
for (i = 0; i < N; i++)
  for (j = 0; j < N; j++)
    sum += A[i][j];
```

Questions:

1. How many `float64` elements fit per cache line?
2. Approximate fraction of accesses that are cache misses?
3. What changes if the loop order is column-wise?

## Solution 6

1. $64/8 = \mathbf{8}$ elements per cache line
2. About $\mathbf{1/8}$ misses and $\mathbf{7/8}$ hits
3. Column-wise access causes many more misses because the accesses become strided instead of contiguous.

# Problem 7

A dataset has 500 million rows with schema:

```
(UserID int64, Timestamp int64, Value float64)
```

Row-store format, no compression. Use $1\ \mathrm{GB} = 10^9$ bytes.

Questions:

1. What is the size of one row in bytes?
2. What is the total file size in GB?
3. A batch gradient descent run reads the full file 10 times across 3 different learning rates. What is the total read I/O in GB?

## Solution 7

1. $8 + 8 + 8 = \mathbf{24}$ bytes
2. $500 \times 10^6 \times 24 = 12 \times 10^9$ bytes $= \mathbf{12}\ \mathrm{GB}$
3. $12 \times 10 \times 3 = \mathbf{360}\ \mathrm{GB}$

# Problem 8

Dataset $D$: $n = 200$ million rows, $d = 4$ numeric features plus a label $Y$, all stored as `float64` in row-store format. Use $1\ \mathrm{GB} = 10^9$ bytes.

Step A: Build a linear model using Batch Gradient Descent. Try 3 learning rates, 4 epochs each.

Step B: Read $D$, append all quadratic pairwise feature interactions (no squared terms), then write the expanded file to disk.

Step C: Rebuild using the same model selection as Step A, but read the expanded file.

Questions:

1. How many new feature columns does Step B add?
2. What is the size of file $D$ in GB?
3. What is the size of the expanded file written in Step B in GB?
4. What is the total I/O cost in GB across all three steps?

# Solution 8

1. $\binom{4}{2} = \mathbf{6}$ new columns
2. $5 \times 8 \times 200$ million $= \mathbf{8}$ GB
3. $(4 + 6 + 1) = 11$ columns, so $11 \times 8 \times 200$ million $= \mathbf{17.6}$ GB
4. Step A: $8 \times 4 \times 3 = 96$ GB

   Step B: $8 + 17.6 = 25.6$ GB

   Step C: $17.6 \times 4 \times 3 = 211.2$ GB

   Total $= 96 + 25.6 + 211.2 = \mathbf{332.8}$ GB

# Problem 9

Assume:

- 24-bit virtual address space
- Page size = 4 KB
- Each page table entry = 4 bytes
- 8 concurrent processes

Questions:

1. How many virtual pages per process?
2. How large is one page table?
3. How much memory is used for all page tables?

# Solution 9

1. $\dfrac{2^{24}}{2^{12}} = 2^{12} = \mathbf{4096}$ pages
2. $4096 \times 4 = 16{,}384$ bytes $= \mathbf{16}\ \mathrm{KB}$
3. $8 \times 16\ \mathrm{KB} = \mathbf{128}\ \mathrm{KB}$

# Problem 10

A process accesses pages:

`ABCADBEABC`

There are 3 physical frames, initially empty.

Questions:

1. How many page faults occur under FIFO?
2. How many page faults occur under LRU?
3. Which policy does better here?

# Solution 10

1. FIFO faults = $\mathbf{8}$
2. LRU faults = $\mathbf{8}$
3. It is a $\mathbf{tie}$ on this trace.

# Problem 11

Three jobs arrive on a single CPU:

| Job | Arrival | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 2 | 4 |
| P3 | 4 | 1 |

Questions:

1. Simulate FCFS and compute average turnaround time.
2. Simulate non-preemptive SJF and compute average turnaround time.
3. Simulate SRTF and compute average turnaround time.

# Solution 11

1. FCFS: `P1 0–7`, `P2 7–11`, `P3 11–12`
   Average turnaround time $= (7 + 9 + 8)/3 = \mathbf{8}$
2. Non-preemptive SJF: `P1 0–7`, `P3 7–8`, `P2 8–12`
   Average turnaround time $= (7 + 4 + 10)/3 = \mathbf{7}$
3. SRTF: `P1 0–2`, `P2 2–4`, `P3 4–5`, `P2 5–7`, `P1 7–12`
   Average turnaround time $= (12 + 5 + 1)/3 = \mathbf{6}$

# Problem 12

A task graph has total work = 100 units and critical path = 40 units.
You have 4 identical workers.

Questions:

1. What is the best possible lower bound on completion time?
2. What is the maximum possible speedup?
3. If a schedule completes in 50 units, what is total idle time?

## Solution 12

1. $\max(100/4, 40) = \mathbf{40}$
2. $100/40 = \mathbf{2.5}\times$
3. $4 \times 50 - 100 = \mathbf{100}$ idle units